

- Q.32 Explain IUPAC naming for functional group like ketone and aldehydes.
- Q.33 Give any five uses of butadiene.
- Q.34 Discuss about any four petrochemicals and their uses.
- Q.35 Explain the significance of IUPAC naming also write IUPAC names of cyclic and aromatic compounds.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is a C₄ compound draw the neat and clean flow sheet of styrene butadiene rubber with process description.
- Q.37 Discuss the manufacturing of vinyl chloride with its properties and flow sheet. Also write any four uses of vinyl chloride.
- Q.38 Explain the manufacturing process of styrene with flow sheet. Also Explain any two major engineering problems for its manufacturing.

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Roll No.

5th Sem / Chem Subject:- Petrochemicals Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is not the product of Ethylene?
- a) Ethyl Benzene b) Cumene
c) Ethanol d) Ethylene Oxide
- Q.2 Which one of the following is a petrochemical?
- a) Acetone b) Coke
c) Paraffin wax d) Ammonia
- Q.3 _____ is used as catalyst in the manufacturing of maleic anhydride from benzene by catalytic vapour phase oxidation
- a) Vanadium pentaoxide
b) Hydrogen peroxide
c) Phosphoric acid
d) Molybdenum naphthanate
- Q.4 Ethylene is produced from naphtha by the process
- a) Catalytic cracking b) Pyrolysis
c) Steam cracking d) None of above

- Q.5 Aromatic compound is _____.
 a) Methanol b) Ethanol
 c) Phenol d) None of the above
- Q.6 The molecular formula of propane is _____.
 a) CH_4 b) C_4H_{10}
 c) C_3H_8 d) C_2H_6
- Q.7 The main use of butadiene is _____.
 a) As a plasticiser for unsaturated polyester
 b) In the manufacture of synthetic rubber.
 c) As an anti-skimming agent in paint.
 d) None of the above
- Q.8 Which of the following is not a property of methanol?
 a) Colourless liquid b) Boiling point 337 K
 c) Immiscible in water d) Highly poisonous
- Q.9 The functional group present in a carboxylic acid is _____.
 a) $-\text{OH}$ b) $-\text{CHO}$
 c) $-\text{COOH}$ d) None of the above
- Q.10 Nail polish remover contains _____.
 a) Petroleum ether b) Benzene
 c) Acetone d) Acetic acid

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the full form of PVC?
 Q.12 List one property of acetone.
 Q.13 Write IUPAC name of C_2H_5 .

- Q.14 Write one example of C_1 compound.
 Q.15 Write two names of petrochemical Industry in India.
 Q.16 Write the formula for styrene.
 Q.17 Name any two aromatic compounds.
 Q.18 Write one use of butadiene.
 Q.19 Write formula for ethylene dichloride.
 Q.20 Write the formula for Alkenes.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write about any five properties and major Engg. Problems of ethylene oxide.
 Q.22 Write five raw materials required for petrochemicals.
 Q.23 Discuss about C_1 compound.
 Q.24 Write five uses of phenol.
 Q.25 Explain about the properties and uses (any five) for acrylonitrile.
 Q.26 Name five chemicals which can be derived from ethylene and state one use of each.
 Q.27 Discuss about the aromatic compounds with examples.
 Q.28 Describe the major engineering problems (any five) for manufacturing of acetone.
 Q.29 What are organic compounds? Also write the IUPAC name of alkanes with examples.
 Q.30 Describe the manufacturing process of cumene.
 Q.31 Draw neat and clean flow sheet of manufacturing of methanol.